Hum 4249 Business Psychology and Communications Credit: 3

Business Psychology: Introduction to Psychology, Psychology in Business; Job Ana Job-oriented Approach, Person-oriented Approach; Assessment Methods for Sele and Placement, Psychological Tests, Training and Development, Theories of Empl Motivation, Job Attitude and Emotion, Productive and Counterproductive Empl Behavior, Occupational Health Psychology, Leadership, Organizational Develop and Theory, Effectiveness of Organizational Development, Socio-technical Sy Theory.

Business Communication: The Role of Communication in Business, Importan Communication Skills, Main Form of Business Communication, Process of Hu Communication, Fundamentals of Business Writing, Basic Pattern of Business sages, Job Search Activities, Fundamentals of Report Writing, Other Forms of Busi Communication.

Recommended Texts:

1. R. V. Lesikar et al., *Basic Business Communications*, 11th ed. McGraw-Hill,
2. P. E. Spector, *Industrial and Organizational Psychology: Research and Pra* 6th ed. Wiley, 2011

Math 4241 Integral Calculus and Differential Equations Credit: 4

Integral Calculus: Definitions of integration, Integration by method of substitutio tegration by the method of successive reduction. Definite integrals. Beta function Gamma function. Area under a plane curve in Cartesian and Polar co-ordinates. of the region enclosed by two curves in Cartesian and Polar co-ordinates, param and pedal equations. Intrinsic equation. Volumes of solids of revolution. Volu hollow solids of revolution. Volume of hollow solids of revolution by shell me Area of surface of revolution.

Ordinary Differential Equation: Degree and order of ordinary differential equat Formation of differential equations. Solutions of first order differential equat by various methods, Solutions of general linear equations of second and hi orders with constant coefficients, Solution of homogeneous linear equations. Sol of differential equations of the higher order when the dependent of indepen variables are absent. Solution of differential equation by the method based o factorization of the operators, Frobenius' method, Bessel's and Legendre's differ equations and polynomials. Partial Differential Equations: Four rules for so simultaneous equations of the form. Lagrange's method of solving PDE of orde Integral surfaces passing through a given curve.

Nonlinear PDE of order one (complete, particular, singular and general integ standard forms $f(p,q)=0, z=px+qy+f(p,q), f(p,q,z)=0, f_1(x,p)=f_2(y$ Charpit's method. Second order PDE: its nomenclature and classifications to cano

(standard) - parabolic, elliptic, hyperbolic. Solution by separation of variables. Linear PDE with constant coefficients.

Recommended Texts:

1. S. L. Ross, *Differential Equations*, 3rd ed. Wiley, 2007
2. E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed. Wiley, 2011
3. E. A. Coddington, *An Introduction to Ordinary Differential Equations*. Dover Publications, 1989

CSE 4203 Discrete Mathematics Credit: 3.00

Set theory, Elementary number theory, Graph theory, Paths and trees, Generating functions, Algebraic structures, Semigraph, Permutation groups, Binary relations, functions, Mathematical logic, Propositional calculus and predicate calculus.

Recommended Texts:

1. K. H. Rosen, *Discrete Mathematics and Its Applications*, 8th ed. McGraw-Hill Education, 2018
2. R. L. Graham et al., *Concrete Mathematics: A Foundation for Computer Science*, 2nd ed. Addison-Wesley Professional, 1994

CSE 4205 Digital Logic Design Credit: 3.00

Number Systems and their conversion, Logic Gates, Boolean algebra, Truth Tables and K Maps, Karnaugh map logic simplification tool, Combinational circuits analysis and design Sequential Circuit Concept: Introduction to Flip-Flops i.e. J-K F/F, Introduction to Latches, design procedures, introduction to developing state diagram and state table, Structured Sequential Circuits: Registers, shift Registers, parallel Loading of Registers, Counters: synchronous, asynchronous, serial Programmable logic: Random access memory (RAM), Programmable logic Array (PLA).

Recommended Texts:

1. M. M. Mano et al., *Logic and Computer Design Fundamentals*, 5th ed. Pearson, 2015
2. B. Holdsworth and C. Woods, *Digital Logic Design*, 4th ed. Elsevier, 2002

SWE 4201 Object Oriented Concepts I Credit: 3.00

Introduction to Object Oriented Concepts: class, object, encapsulation, inheritance, interfaces, Using UML to model a Class Diagram, Constructor, Polymorphism, Aggregation and Composition, Error handling and The concept of scope;